

Applicant Information

Tracking Number: GRANT14394092**Application #: 1949260****Date: 04-03-2025 23:49 PM**

Grants.gov Information

Category	Research
Opportunity	
Name	CymaSpace
Project Start Date	01-01-2026
Project End Date	12-31-2027

Organization Details

NEA Discipline for Proposed Project

Research

Legal/IRS Name (per your IRS Determination Letter) *

CymaSpace

Popular Name (if different)**Mission of Your Organization: ***

CymaSpace creates cultural equity for our communities by advocating for accessible and inclusive Art, STEM, Education and Workforce Development.

Organizational Background *

CymaSpace is a nonprofit, Deaf-led Disability Arts & Cultural Hub in Portland, Oregon with the mission of making arts, culture and media accessible to the Deaf community through Technology, Education and Outreach. We promote Deaf performers, train media technicians, mentor Disability Artists, and enable the diverse cultural communities that intersect with Deafness to enjoy representation within our organization and at public events throughout Portland and beyond.

CymaSpace is inspired by the opportunities that cymatics, the study of translating sound waves into visual and tactile phenomena, provides to improve experiences for Deaf, DeafBlind, Hard of Hearing and hearing people. We produce creative technology-driven media for live performances, artistic installations, and video recordings to “make sound visible.” Physically, we enable access through Open Captioning displays, sound activated light systems, and innovative multimedia art. Emotionally, we create opportunities for participants to feel deeply connected to works of

art and communal activities that traditionally exclude them. Our work bridges sound—spoken words, sound effects and music—with its sensory partners—written text, light, vibrations, and patterns—to create experiences that resonate with all participants. CymaSpace advocates for the Deaf community by partnering with local and national art, science and cultural organizations. Coordination of sign language services and open-captioned technical support provides Deaf people with communication access. We believe in creating an environment of Equity that addresses the needs of Deaf people. We operate among the intersections of Deafness with other cultures and disability groups. We build community between disabled and non-disabled attendees and collaborators.

Organization Budget

	Most Recently Completed FY	Previous FY	Two FYs Prior
Fiscal Year End Date (MM/DD/YYYY)	12/31/2024	12/31/2023	12/31/2022
Total Income	\$332,347	\$562,466	\$132,616
Total Expenses	\$563,150	\$359,022	\$133,426
Operating Surplus and Deficit	(\$230,803)	\$203,444	(\$810)

Fiscal Health: *

In the past three years, we have significantly and rapidly increased revenues from foundation and government grants. Our largest jump was from 2022 to 2023 when we received a large \$206,083 grant from the Mt. Hood Cable Regulatory Commission (MHCRC), primarily to purchase equipment and hire an equipment manager to identify and purchase this equipment. Our annual deficits typically total less than \$1,000, and we experience a significant surplus in 2023. In December 2024, a deficit was caused by the end of funding by a previously reliable funder as well as a major transition moving locations and opening a new community space and cafe to increase our capacity to host in-person events in Fall 2024. We have increased our ability to generate unrestricted earned income through our new cafe while we continue to apply for grants and begin development of a formal individual donor program. The Collins Foundation has recommended us for unrestricted emergency funding in 2025.

Operating History

Year 1 *

2020

**Representative
Example Year 1 ***

- * 2020 - Present (ongoing)
- * Audiolux Audio Reactive Lights Research
- * AudioLux is a product that visualizes music or other performing arts on LED strips. Oregon State University Engineering Majors Capstone project with ongoing development and refinement.
- * Oregon State University, Corvallis, Oregon
- * Dr. Chet Udell (supervisor), Myles de Baston (IP Owner/Advisor), Longjie Guan, John Zontos, Ming Wei, Evan Cochran
- * 2020
- * Artist in Residence
- * Funded by Andy Warhol Foundation to produce 3,000 sq ft experiential site installation and performances with universal design and accessibility features.
- * Portland Institute for Contemporary Art, Portland, OR
- * Myles de Bastion
- * 2020
- * Artist in Residence
- * Recorded an album of self-produced music with Chris Funk (The Decemberists), featuring Vaughn Brown a DeafBlind drummer culminating in a performance of music and light in their skateboard bowl.
- * Portland, OR
- * Myles de Bastion, Chris Funk, Vaughn Brown

Year 2 *

**Representative
Example Year 2 ***

- 2021
- * 2021 - Present (ongoing)
 - * Vibrosonics haptic for music research
 - * Provide audio-based entertainment for Deaf/Hard-of-Hearing audiences and enhance audio-related experiences through touch sensations. Oregon State University Engineering Majors Capstone.
 - * Oregon State University, Corvallis, Oregon
 - * Myles de Bastion, Nick Synytsia, Alex Synytsia, Vincent Vaughn, Dr. Chet Udell
 - * 2021 - 2024
 - * Sign Language Recognition in Virtual Reality
 - * Integrate sign language recognition into virtual reality environments
 - * Gallaudet University, Washington DC
 - * Myles de Bastion

Year 3 *

**Representative
Example Year 3 ***

- 2022
- * 2022
 - * CymaSpace Celebration Concert/Guthman
 - * Live performance featuring multi-sensory instruments to interpret music through light and touch using novel and experimental equipment and instruments.
 - * Physical and Zoom, CymaSpace, Portland, Oregon
 - * Chet Udell, Myles de Bastion, Nate Hergert, Dillon Simeone

Year 4 *

2023

Representative

* 2023

Example Year 4 *

* Haptic Dance Floor (Oregon Creative Foundation Creative Heights Award)

* Using Universal Design and haptics vibration technology to build a dance floor and bodysuits that translate music from an auditory experience to a physical one

* Portland, OR

* Nate Hergert, Shawn Trail, Dillon Simeone, Myles de Bastion, Chet Udell

Year 5 *

2024

Representative

* 2024

Example Year 5 *

* ASL champ! Sign Language Virtual Reality Game

* Development of a fully functional ASL learning game in VR. The game uses an interactive approach where users learn ASL signs from a signing avatar.

* Gallaudet University, Washington DC

* Md Shahinur Alam, Jason Lamberton, Jianye Wang, Carly Leannah, Sarah Miller, Joseph Palagano, Myles de Bastion, Heather L. Smith, Melissa Malzkuhn, Lorna C. Quandt

* 2024

* GestoLumina: Gesture interpreted Light, Sound and Haptics. Towards a Framework for Universal Music Design

* Creation of a modular interface to allow user to choose between visual and/or haptic feedback along with gestural sensory utilizing two sensory bracelets and an audio-reactive LED color feedback array.

* Remote collaboration, research conference publication: New Interfaces for Musical Expression 2024. Utrecht, Netherlands

* Nate Hergert, Dillon Simeone, Doga Cavdir, Duncan MacConnell, Shawn Trail, Myles de Bastion

Project Details

**Project
Synopsis ***

To create a music-focused STEAM curriculum for Deaf and Hard-of-Hearing K-12 youth.

**Research
Agenda Topic
Area ***

Measuring the impacts of the arts

Select the category that best fits your project from the NEA priority research topic areas below.

Project Summary *

On Cognition and Learning

Project
Description *

a. RESEARCH MOTIVATION.

Research Question: How does a Deaf-centered STEAM music curriculum improve engagement, learning outcomes, and creative expression for K-12 students, particularly those who are Deaf or Hard-of-Hearing?

Music has an indelible presence, shaping emotions, culture, and experiences. However, for the Deaf and Hard-of-Hearing (DHH), this profound avenue of human expression remains largely inaccessible. Music serves as a conduit for emotional resonance and social connection within the hearing-able world, affording music-makers and audiences with sonic agency [1]. Sonic Agency is the foundation of CymaSpace's Universal Music Design (UMD) project and transcends auditory perception, advocating for the expansion of avenues through which all individuals can engage with and experience music's profound impact on the human psyche [2]. This project aims to bridge the auditory richness of music with the tactile world, using assistive technologies such as haptic vibration and immersive audio reactive lighting that translates audio frequencies into color. These innovations provide a novel means to tangibly experience the essence of music. Early music training helps children develop equilibrium and proprioception, but DHH children do not develop these skills in the same manner as a hearing-able child, and can struggle with activities requiring balance, such as riding a bicycle [15].

In the K-12 school setting, despite facing marginalization from music education activities [20], DHH music students succeed at learning music when given the opportunity [23]. Additionally, accessibility research has an under-representation of DHH designers, musicians, composers, and performers as active participants throughout the design research phases [3]. This project will engage collaborators from DHH and hearing populations and Cochlear Implant/Hearing Aid (CI/HA) users.

Project Goals

Investigate the Impact of Multisensory Music Education:

- * Conduct laboratory experiments to determine optimal multisensory configurations (Months 1-6)
- * Test cochlear implant programming optimizations for music perception (Months 1-6)
- * Measure how students engage with music through tactile, visual, and interactive technologies in classroom settings (Months 7-12)

Develop and Test Evidence-Based Curriculum Model:

- * Design K-12 learning modules concurrently with perceptual study findings (Months 1-6)
- * Incorporate optimal multisensory configurations identified in lab testing
- * Pilot these modules integrating vibration, visual music notation, and STEAM experiments (Months 7-12)

Begin Assessment of Educational and Social Outcomes:

- * Study initial changes in student confidence, creative participation, and cross-

disciplinary learning

- * Document educator observations on classroom dynamics and inclusive practices
- * Compare classroom outcomes with laboratory findings

This study seeks to answer the following research questions: (1) What educational tools can be developed to increase multi-sensory music education in K-12 settings? (2) What are DHH student perceptions of their multisensory musical experiences of a haptic curricular prototype? (3) What are DHH students' attitudes towards the use of the haptic curricular prototype in their multisensory musical experiences? (4) Does a multi-sensory haptic curriculum improve music knowledge and understanding for K-12 DHH students?

Through early phases of this investigation, UMD Team Leader, Nate Hergert (Deaf/CI user) and UMD Lead Deaf Design Engineer Dillon Simeone have developed and tested haptic technologies training themselves in music theory and audio engineering concepts towards arriving at a platform for experimentation. As this work requires a controlled acoustical environment, Hergert consulted with an acoustical architect towards the acoustical analysis and design of the studio purpose built for DHH audio and haptics research.

The team has since developed a custom framework with novel interfaces and haptics for Deaf music creation. Our first interface GestoLumina (GeLu) builds directly on the work of Advisory Board member Doga Cavdir [25], while incorporating the UMD team's background in both complex non-invasive, idiomatic 3D musical gesture tracking [26] and ASL gesture recognition [27]. This study will be modeled after "Sonic Interaction Design" (MIT Press) [35], advised by Board Member Stefania Serafin, Ph.D., (editor of "Sonic Interaction Design").

b. RESEARCH DESIGN.

We will develop a curricular prototype to promote sonic agency and improve music education experiences and outcomes for K-12 DHH students. This curriculum will incorporate haptic vibration devices to translate sound for DHH students and will be tested for impact on their ability to engage with a music education curriculum. This qualitative study involves a triangulation of approaches including surveys, interviews, and observations in both a laboratory and practical setting.

Based on early exploratory workshops with DHH students, it is anticipated that the results of this study will provide valuable information for engineers on the efficacy of the designed haptic devices incorporated into the curricular approach for including DHH students with multisensory musical experiences in the school classroom [64] as well as inform educators of the DHH in specialist and mainstream schools on ways to include DHH students in a music-centric STEM education [65], with implications for curriculum design and implementation [66].

Methodology

- * Concurrent Laboratory Research and Curriculum Design (Months 1-6)
- * Assess the status of available technologies, including technologies developed by the project team.
- * Conduct a thorough literature review of available papers on music education

curricula engaging universal design and relating that to state and national educational learning standards

- * Verify or complete certification for all participating research interviewers
- * Partner with audiology experts to study perception of musical attributes under different sensory conditions
- * Apply emerging findings directly to curriculum design process
- * Develop learning modules that incorporate optimal sensory configurations

Implement Evidence-Based Learning Modules (Months 7-12)

- * Deploy modules in partner schools with diverse student populations, including DHH learners
- * Ensure classroom activities reflect laboratory-validated approaches

Collect Comprehensive Data (Full Year)

- * Controlled laboratory experiments (Months 1-6)
- * Classroom observations using structured protocols (Months 7-12)
- * Student feedback surveys (adapted for accessibility needs)
- * Educator assessment interviews
- * Documentation of student work and creative responses

Analyze Results to determine:

- * Correlation between laboratory findings and classroom effectiveness
- * Most effective multisensory combinations for different musical attributes
- * Impact of optimized cochlear implant programming on music appreciation
- * Areas for curriculum refinement in potential subsequent years

We will assess the status and impact of available technologies on DHH student learning, including technologies developed by the project team through grants from Oregon Community Foundation and Oregon Arts Commission. We will also collaborate with academic and scientific advisors to create a music curriculum informed by Universal Design that incorporates a prototype for haptic-based music education.

The UMD Team is composed of NEA Research Program Director, Shawn Trail, PhD (CymasSpace Research & Production Coordinator, Music Technology Pedagogy Pioneer); UMD Program Director, Nate Hergert (Deaf/CI User, Music Producer and DJ, DHH Access/Inclusion Activist, Educator); and UMD Lead Design Engineer, Dillon Simeon (DHH assistive technology HW/SW developer, Deaf Music Student). Our Advisory Board is composed of academic and research experts.

This project will utilize tools developed by project partners, including a guitar framework that allows a Deaf/Blind guitarist to improvise with other musicians, GestoLumina (GeLu), an American Sign Language music platform, a Xylophone Framework inspired by the Orff child-centered learning model, and the Common Affordable 4 Channel Wireless Vibration Capture System.

The R&D will inform the design of an experimental DHH STEAM curriculum with Cymaspace technologies reflected in a novel inclusive networked classroom model [51]. Design of evaluation methods for auditory perception, as well as user interfaces will be supervised by Doga Cavdir, building on work investigating multi-sensory design and equitable concert spaces for CI users [57], [79].

The evaluation of novel DHH music user interfaces [56], auditory scope of perception of CI users towards music production [57], and curriculum [58] will involve specialists represented in our Advisory Board, participants/volunteers from the DHH community, audiologists, partner organizations, industry partners, higher learning education programs, and the educators from the public school school system.

I. Sample(s).

Our control group will consist of: Nate Principal Investigator (Cochlear Implant User), Myles (Hearing Aid User), Shawn (Hearing), and Dillon (totally Deaf). Inclusion criteria for research group students include proficiency in reading and writing English and fluency in American Sign Language (ASL). Exclusion criteria involve any medical condition in which multisensory stimulation might lead to distress in the student, such as hypersensitivity. We expect to study a maximum of 50 students across all programs.

Participants in the study will engage in up to three test occurrences over up to three days. Each test occurrence will include: one standard curricular music education experience, such as a demonstration of music scales on a piano or guitar, without intervention; the same music education experience with the intervention of engineer-designed haptic devices [84]; followed by one of three research instruments. Each occurrence will take no longer than the length of a typical classroom lesson (up to 60 minutes). In total, each participating student will participate in up to three test occurrences.

Researchers will collect three sets of information from each student. First, the researchers will conduct a survey with open-ended questions, where students will record their perceptions of their musical experience as impacted by the inclusion of specific haptic devices in classroom music education activities. Questions will be written in a style that is appropriate for participants 13 and older. Students will be given appropriate lengths of time to complete their responses to each activity. The overall experience will be designed to be completed within one typical lesson period of the normal school day. Second, researchers will work with students individually in a smaller office space, on a different day, where similar activities will be offered and field notes will be taken by the researcher to record observed behaviors. Third, researchers will conduct scripted interviews with the same students individually using open-ended questions. Questions will again be designed in a style that is appropriate to the age group of the participants. The overall experience of the observation and interview will be designed to be completed within one typical lesson period of the normal school day. All three sets

of data will be coded to preserve anonymity and will be analyzed for differences in student perception of classroom musical experiences, with and without the haptic devices.

Research Procedure: By mutual agreement between the school administrators, relevant faculty, and the researcher, an oral presentation will be given to students introducing them to the study. Parental consent and student assent will be obtained in writing on the form prior to the start of the study. Students meeting inclusion criteria will be presented with a range of educational musical experiences. The researcher will take field notes on observed student responses and will follow with the interview, asking students scripted questions about their experiences that will aim to obtain more detailed information than was achieved in the survey alone.

II. Data Source(s).

Data collection will be accomplished through three sources: surveys written by students at the Oregon School for the Deaf or other participating institution(s), researcher field notes written during observations of the same students, and transcripts of audio recordings of student responses to interviews. Because audio recordings of the interview responses will have identifying voice recognition data, the recordings will be destroyed immediately after transcription. Data will be coded to respect anonymity of students through data anonymization, randomization, and masking using Python using an in-house engineer. Data will be stored on the researcher's encrypted computer that is password protected. Data will be destroyed within five years after the study is complete. The data is limited to this specific sample of participants. It is further limited by the capacity of the students to express the nature of their experiences in written and oral communications. Balance to the self-reporting will be achieved through the observations of facial and gestural behaviors of students by the researcher during the musical experiences with and without the haptic devices. There will be a pause after each experience to allow the researcher to make notes. The researcher will follow a study-specific protocol to remain objective and reduce bias while noting student responses during observations. When participants are recruited, and when the study is administered, the importance of accuracy in responses will be explained as well as the fact of there being no right, wrong, or preferred answers. While each new experience will be presented both with and without the haptic devices, the inclusion or exclusion of the haptic device will occur in random order each time.

These data will inform the impact that haptic device interventions have on DHH students when learning music. Anticipated limitations posed by the data include a reliance on DHH students fluent in ASL who can also read and write in an additional language, English. This may bias the sample toward DHH students with greater exposure and integration in the hearing world than DHH students who primarily operate within the Deaf culture.

Routine monitoring and oversight includes quarterly reporting to and review by our expert Advisory Board, with increased oversight occurring in the weeks prior to, during, and following our research with live research subjects.

The identified sample size will represent the greater of 10 DHH students or 10% of the DHH students who are 13 or older within the populations studied.

c. DATA ANALYSIS.

Once surveys, field notes, and interview transcripts have been collected, the analysis of qualitative data will commence. Data will be coded and entered into a computer database. All coding of data and allocation of data categories will follow an emergent design because the researcher expects to see unanticipated answers and views expressed by the students. All data will be analyzed for trends and discrepancies in the results. Each researcher will write a narrative report, and a discussion of those results will follow. Assuming a methodological assumption, categories will be assigned during the analysis of data to allow for new categories to emerge that might not have been envisaged in advance.

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Data Access Plan *

Project partners will operate within the guidelines of their respective Data Access Plans, and Advisory Board members will provide oversight of the implementation of proposed data collection to ensure compliance with all relevant laws and regulations.

OHSU, OSU and CRIS all have rigorous administrative protocols and/or an Institutional Review Board to which relevant human subject research proposals will be submitted prior to initiation of the research. Under the directive of the research partners at all participating institutions, CymaSpace will be authorized to collect/access data specified in the application.

We will recruit participants, including DHH students, directly through our partner Columbia Regional Inclusive Services ensuring access to the required student data, including classroom observations, feedback surveys, and educational assessments. This data will be used to evaluate the effectiveness of a Deaf-centered STEAM music curriculum.

Evidence of Public Domain Data: We will use publicly available data on multisensory learning and educational outcomes in music education to complement primary data collection. This may include existing studies, educational models, and research on music perception and cognition in the DHH community. Our research findings will contribute to this body of knowledge once published.

Process for Obtaining External Data: For secondary data not collected within the scope of this project, we will acquire access through formal partnerships with institutions that collect relevant data on student outcomes, multisensory learning, or music education. The process will include:

- * Initial Contact and Agreement: We will reach out to relevant institutions, outlining our research goals to ensure mutual alignment of interests, sign data-sharing agreements, and secure necessary permissions.
- * Timeline for Access: Data-sharing agreements will be finalized by February 1, 2026, ensuring access rights by the end of Month 2 of the project.
- * Cost and Restrictions: We anticipate no significant cost or restrictions on accessing the data, although confidentiality or time-related conditions imposed by data providers will be adhered to.

Institutional Review Board (IRB) Approval and Ethics Training: As part of our commitment to ethical research, we will submit our research proposal to the IRB of our partner institutions for approval before starting data collection, ensuring compliance with federal and institutional guidelines, particularly for vulnerable populations like students with disabilities.

* IRB Approval Process: We have initiated the IRB review process and expect to receive approval before the start of the grant timeline, after finalizing our recruitment strategy and informed consent procedures. The IRB will also review our data collection tools (surveys, assessments) to ensure ethical and non-invasive methods.

* Ethics Training for Key Personnel: All researchers involved in primary data

collection and analysis will complete ethics training on human subjects protections. We are committed to ensuring all team members are educated on ethical research, particularly regarding the collection of sensitive data from minors and individuals with disabilities. Certificates of completion for all key personnel will be obtained before initiating data collection and submitted to the NEA when required.

Data Collection and Security Measures:

* Data Collection Methods: Data will be collected through classroom observations, structured surveys, interviews with educators, and student work samples. All data will be anonymized to protect student identities, and personally identifiable information will be securely stored and accessible only to authorized staff.

* Data Security: We will implement robust data security measures, including encrypted digital storage, secure access protocols for data analysis, and a clear data disposal policy at the conclusion of the project. Data shared with external researchers will comply with all data-sharing agreements and confidentiality requirements.

Dissemination of Research Results: Upon completing data collection and analysis, we will disseminate findings through peer-reviewed journals, educational conferences, and workshops. We will collaborate with educational resource publishers to create tools for teachers based on the research findings. These resources will include modules and guidelines on integrating multisensory music education in inclusive classrooms. All research outcomes will be made publicly available to contribute to the broader field of Deaf education and music pedagogy.

Conclusion: We are committed to conducting research that is ethically sound and produces reliable data on the impact of multisensory music education for DHH students. Our team is prepared to adhere to all IRB and ethical guidelines, ensuring the safety and dignity of our participants. We will manage the data secure

Data Management Plan *

Data Collection Framework

Laboratory Research Data (Months 1-6)

- * Quantitative performance metrics across different sensory conditions
- * Comparative analysis of musical attribute perception (pitch, timbre, rhythm)
- * Cochlear implant filterbank optimization results
- * Participant feedback on different multisensory configurations

Curriculum Design Process Data (Months 1-6)

- * Documentation of how laboratory findings inform specific module components
- * Prototype testing results with small student groups
- * Iterative design modifications based on emerging research findings

Curriculum Implementation Data (Months 7-12)

- * Pre/Post Assessments: Measure changes in understanding of sound concepts, music appreciation, and STEAM connections
- * Creative Portfolio Development: Document student creative work throughout the modules
- * Participation Metrics: Track engagement levels, time on task, and voluntary participation
- * Student Reflections: Gather feedback on accessibility, enjoyment, and learning impact

Data collection will be accomplished through five sources: audiology tests at OHSU, perceptual studies at Telegraph Mastering, surveys written by students at Creston Public School (or other participating institution(s) where the DHH student cohort attends with support via CRIS, researcher field notes written during observations of the same students, and transcripts of audio recordings of student responses to interviews.

Collected data will include participant information, including age, grade, and hearing ability. Each subject will be assigned a code to anonymize participant identities in written documentation. Collected data will also include participant responses to the music education experiences presented, including perceived engagement and retention (as self-identified by the student), written and audio documentation of verbal and ASL signed responses to interview questions, and researcher observations of student behaviors.

Data will be stored on the researcher's encrypted computer that is password protected. The data is limited to this specific sample of participants. It is further limited by the capacity of the students to express the nature of their experiences in written and oral communications. Balance to the self-reporting will be achieved through the observations of facial and gestural behaviors of students by the researcher during the musical experiences with and without the haptic devices. There will be a pause after each experience to allow the researcher to make notes. The researcher will follow a study-specific protocol to remain objective and reduce bias while noting student responses during observations. When participants are recruited, and when the study is administered, the importance of accuracy in responses will be explained as well as the fact of there being no right, wrong, or

preferred answers. While each new experience will be presented both with and without the haptic devices, the inclusion or exclusion of the haptic device will occur in random order each time.

Data will be collected and analyzed, with the accumulated results culled into a report, research publications, a book, and a web repository. The report will be made available as Open Source, Public Data that leading to peer reviewed publications.

Because audio recordings of the interview responses will have identifying voice recognition data, the recordings will be destroyed immediately after transcription. Individual subject data and observations will be destroyed within five years after the study is complete.

Year-End Deliverables

- * Comprehensive analysis of laboratory research findings
- * Documentation of research-informed curriculum development process
- * Implementation results and student impact assessment
- * Recommendations for curriculum refinement and scalability
- * Research report connecting laboratory findings to classroom outcomes
- * Potential recommendations for cochlear implant programming to enhance music appreciation
- * Multiple journal/conference publication submissions

Project Partners

Selection of Key Organizational Partners *

Key organizational partners were selected to represent a balanced team of arts and research partners, including representation of project leads experiencing Deafness or hearing loss and project partners working directly with the target K-12 youth population. The five partners listed have already engaged with project leadership to conceptualize this proposal and to assist with the implementation of preliminary music education workshops to gain qualitative insights about student engagement and the applicability of assorted music instruments and equipment in a classroom setting.

Additional project partners include Oregon School of the Deaf, Woojer (a music instrument manufacturer), Portland Community College, and Not Impossible Labs.

Description Of Organizational Partner:

Organization Name: *

Oregon State University

Proposed or committed? *

Committed

Organization Type *

Other College/University

Organization Description *

The OPEnS Lab expands the possibilities of scientific observation by combining open-source development and cutting edge technology- primarily developing environmental sensing projects and research. Our lab provides the tools necessary for students, professors, and researchers alike to develop tools used locally and worldwide by the ecological science and engineering community. OPEnS will be providing custom multi-modal technology solutions for DHH music curriculum design and experimentation.

Organization Name: *

Columbia Regional Inclusive Services (CRIS)/Portland Public

Proposed or committed? *

Committed

Organization Type *

K-12 School/School District

Organization Description *

CRIS provides services under the Oregon Department of Education's regional program to support and fund special education. Offerings include parent consultation, early intervention, school-age support, itinerant teaching, physical therapy, assistive technology, augmentative communication, and Deaf and Hard-of-Hearing classrooms. Staff are certified and experienced in current trends, research, and technology to deliver specialized, high-quality education.

Organization Name: *

Oregon Health and Science University (OHSU) Cochlear Implant

Proposed or committed? *

Committed

Organization Type *

Other College/University

Organization Description *

The Reiss Lab studies auditory perception and how this is affected by hearing loss, with a focus on binaural fusion and spectral integration. The Lab also studies how auditory nerve function changes over time after cochlear implantation in both humans and animal models. The Lab has also been involved for over 25 years with HI-ARO (hearing-impaired Association for Research in Otolaryngology), a mentoring network for scientists, engineers, and clinicians with hearing loss in the field of hearing

Organization Name: *

Eastern Washington University

Proposed or committed? *

Committed

Organization Type *

Other College/University

Organization Description *

Founded in 1882, Eastern Washington University (EWU) is the region's polytechnic school. EWU provides an education rooted in the liberal arts, designed to address the unique challenges and opportunities facing the region and we are collaborating with EWU's Director of Music Education to build out our music curriculum.

Organization Name: *

Telegraph Mastering

Proposed or committed? *

Committed

Organization Type *

Other; please specify:

Please specify other:

For-Profit Business

Organization Description *

Telegraph Mastering is a full-service mastering studio located in Portland, OR offering mastering for CD, download, and vinyl with in-house lacquer cutting and will provide support for mastering audio files from our music curriculum.

Key Individuals

Selection of Key Individuals *

Key Individuals reflect the multidisciplinary nature of the project scope. Universal Music Design (UMD) Team Leader, Nate Hergert (Deaf cochlear implant user) developed and tested haptic technologies to reflect music theory and music technology concepts using a controlled acoustical environment. Hergert and Dr. Shawn Trail then consulted with an acoustical architect to analyze and fully acoustically treat the Universal Music Design program studio, leading to Hergert likely becoming one of the world's first Deaf studio acoustical designers. Dr. Lina Reiss provides deep experience with conducting perceptual studies with cochlear implant users, and has made meaningful contributions to the project proposal. Dr. Chet Udell is a long-time CymaSpace collaborator, creating musical instruments that interpret music into visible light. Finally, Kristy Karsten provides deep knowledge of working with DHH youth in educational settings, guiding curriculum development.

Last Name: *

Trail

First Name: *

Shawn

Proposed or committed? *

Committed

Bio including the proposed role in the project *

Dr. Shawn Trail (PROJECT DIRECTOR) is a musician, technologist, and researcher whose work bridges cutting-edge music technology, accessibility, and professional practice. With collaborations alongside Grammy-winning and Multi-Platinum artists, Trail brings a rare mix of creative vision and technical depth to inclusive innovation. Earning a PhD in computer science applied to music, Trail focused on assistive music performance and accessible design. As a neurodivergent scholar (autistic/ADHD/OCD), his perspective drives equity-centered solutions. Through CymaSpace, Trail develops multimodal, haptic systems that make music more accessible for Deaf and Hard-of-Hearing (DHH) communities. A Fulbright Fellow in Stefania Serafin's impactful disability UX lab and PhD mentee of AI audio pioneer George Tzanetakis, Trail has developed dynamic leadership skills for interdisciplinary teams. His work spans research, industry, and pedagogy—pushing the boundaries of technology.

Last Name: *

Reiss

First Name: *

Lina

Proposed or committed? *

Committed

Bio including the proposed role in the project *

Dr. Lina Reiss (PROFESSOR/RESEARCHER) received her Ph.D. in Biomedical Engineering from the Johns Hopkins University in 2005, with a focus on neurophysiology of sound localization circuits in the auditory brainstem. She holds a B.S.E. in Mechanical Engineering from Princeton University. She spent five years as a postdoctoral fellow in the area of cochlear implants and psychoacoustics at the University of Iowa. In 2010, she was appointed as faculty in the Department of Otolaryngology and the Oregon Hearing Research Center at Oregon Health and Science University, with joint appointments in the Departments of Biomedical Engineering and Behavioral Neuroscience. She conducts research on auditory perception in people with hearing loss and cochlear implants. As Founding Director of the Reiss Lab at OHSU, she is a Professor of Otolaryngology-HNS, Biomedical Engineering, and Behavioral Neuroscience and will provide methodology and laboratory support to conduct scientific audiology testing.

Last Name: *

Udell

First Name: *

Chet

Proposed or committed? *

Committed

Bio including the proposed role in the project *

Dr. Chet Udell (PROFESSOR/ENGINEER) is an Associate Professor of Biological and Ecological Engineering at Oregon State University and the Multidisciplinary Programs Administrator for the Design for Social Impact certificate. He directs the OPEnS Lab, a globally recognized NSF- and USDA-funded makerspace with over 30 undergraduate researchers from diverse engineering and science disciplines. His work advances scientific instrumentation for climate, environmental, and agricultural research. A longtime collaborator with CymaSpace, he has performed and conducted workshops globally, integrating new musical instruments and multisensory STEAM education tools. His partnerships include the How Big is Your Dream Foundation, Georgia Institute of Technology, and the Intel-4H-OSU Mariachi STEAM Camp. In this project, he brings deep experience in sensor design, haptics, and student-led innovation to help create inclusive, multisensory music learning experiences for Deaf and Hard-of-Hearing students.

Last Name: *

Karsten

First Name: *

Kristy

Proposed or committed? *

Committed

Bio including the proposed role in the project *

Kristy Karsten is the Program Administrator for Columbia Regional Inclusive Services for the Deaf and Hard of Hearing, a role she has held for a decade. She leads initiatives that provide specialized resources and support to approximately 500 Deaf and Hard-of-Hearing (DHH) students from birth to age 21, ensuring equitable access to high-quality education. A former Literacy Specialist and classroom teacher, Kristy's dedication to DHH education began when her niece was born Deaf during her senior year of high school. Since then, she has been a steadfast advocate for inclusive learning, seeking out and implementing curricula designed by and for the Deaf community. Kristy actively collaborates with educators, families, and Deaf leaders to create culturally and linguistically responsive learning environments. Her leadership ensures that students receive innovative, research-based instruction that honors their identities while fostering academic and personal success.

Last Name: *

Hergert

First Name: *

Nate

Proposed or committed? *

Committed

Bio including the proposed role in the project *

?Nate Hergert (PRINCIPAL INVESTIGATOR) is the Program Director for Universal Music Design (UMD) at CymaSpace, leading initiatives to enhance music accessibility for the Deaf and Hard of Hearing (DHH) community. As a Deaf DJ and Music Producer, Nate combines his passion for music with a commitment to inclusivity. He co-founded Confluence Recording Company, one of the world's first DHH focused music production research studios to develop innovative technologies, translating music into tactile experiences. His work includes the creation of "GestoLumina", an interface which combines gesture sensing and haptic feedback to provide immersive musical experiences. Nate's leadership extends to community engagement, exemplified by organizing events such as the Deaf Holidays party in Portland, Oregon, fostering connections among DHH individuals. His entrepreneurial spirit and dedication to research and development drives UMD's mission.

Other Project Details

Project Start Date *	1/1/2026
Project End Date *	12/31/2026
Promotion & Publicity *	Columbia Regional Inclusive Services will assist with working with their student population, ensuring diversity across student populations, including those who are DHH. Recruitment efforts will include working closely with school administrators, educators, and audiology experts to identify potential participants and gain parental consent. Data collection will involve classroom observations, student feedback surveys, and educator assessments to measure engagement and learning outcomes. Research findings will be made open source and shared widely through publications in peer-reviewed scientific journals and educational learning resource outlets. This will ensure that both academic and practical insights on multisensory music education are accessible to educators, researchers, and the broader public, supporting the advancement of inclusive, Deaf-centered teaching practices in STEAM.
Schedule of Key Project Dates *	Building upon preliminary research conducted by the project team, this one year research project will begin January 1, 2026 and conclude December 31, 2026. PHASE 1 (Jan - June 2026) January: Host Advisory Board Meeting. Assess the status of available technologies, including technologies developed by the project team through a grant from the Oregon Community Foundation and Oregon Arts Commission. Conduct a thorough literature review of available papers on music education curricula engaging universal design and relating that to state and national educational learning standards. February - July: Verify or complete certification for all participating research interviewers. Prepare a perceptual study with audiology lab partners while concurrently designing learning modules. Host additional Advisory Board Meeting in April. PHASE 2 (July - December 2026) July - October: Host Advisory Board Meetings in July and October. Implement and assess pilot lessons given by teachers to classes that include DHH students, using materials and technology designed by the group. November - December: Generate and disseminate reports.

Performance
Monitoring *

The Curricular effectiveness assessment will be achieved through qualitative surveys and interviews with teachers and students, triangulated with observations of the classes by field researchers. Results will indicate measures of: (1) achievement of student learning outcomes, (2) student and teacher perceptions regarding approaches, methods, materials, and technology used during the lessons, (3) reflection on impact of the experiences in the lessons, and (4) ideas for improving the lesson experience. These investigations will be designed to inform the development of robust lesson plans and materials for the implementation phase in the second half of the year.

Engagement
With Intended
Community,
Participants &
Audience *

Proposed work will engage both Deaf and hearing collaborators, with an emphasis on Deaf and Hard-of-Hearing (DHH) researchers and program participants. The project team largely represents the intended community to be served and has been collaborating with many of our key collaborators and intended communities for years.

The Universal Music Design curriculum will ultimately serve DHH youth in grades K-12, so we will host workshops with DHH youth in Elementary School (K-5), Middle School (6-8), and High School (9-12). We are partnering with Columbia Regional Inclusive Services (CRIS), the Portland Public School System, and Oregon School of the Deaf to engage with both DHH educators and DHH students in grades K-12. Project Leads will coordinate with these partners to recruit project participants. We have already had the opportunity to host a test workshop led by Nate Hergert to gain qualitative insights on DHH student engagement with the test curriculum.

In addition, we will utilize the existing networks of CymaSpace and project partners to disseminate results of this work and to recruit participants for future phases of this project.

Project Budget

Project Expenses

DIRECT COSTS: SALARIES & WAGES				
Title and/or Type of Personnel	Number of personnel	Annual or Average Salary Range	% of time devoted to this project	Amount
Executive Director	1	\$ 4,800/year	25%	\$1,200
Bookkeeper	1	\$ 5,200/year	23%	\$1,200
		\$		

		\$				
		\$				
		\$				
		\$				
<u>Total Salaries and Wages</u>				\$2,400		
FRINGE BENEFITS	\$0					
<u>Total Salaries and Fringe Benefits</u>	\$2,400					
DIRECT COSTS: TRAVEL						
# of travelers	From	To	Amount			
<u>Total Travel Expenses</u>			\$0			
DIRECT COSTS: OTHER						
Description	Amount					
Subcontractors (Lead Researchers = \$28K/yr x 3 people)	\$84,000					
Subject Matter Experts in Curriculum Design and Technology Design = \$76,800 (95 hours x \$200/hr x 4)	\$76,800					
Audio Engineers (Digital Assets Manager = \$2,400 @ \$200/day for 12 days; Audio Mastering = \$1K)	\$3,400					

Research Audiologist (80 hours @ \$62.50/hr)	\$5,000
Space/facility/venue costs	\$18,672
Equipment	\$10,000
Supplies/Materials	\$1,500
Training (IRB Training = \$300/person x 2)	\$600
American Sign Language Interpreters	\$1,150
Legal (legal and patent fees)	\$1,500
Utilities	\$2,400
Honoraria for Research Participants (\$25/hr for participants w cochlear implants; \$15/hr hearing)	\$200
	\$0
	\$0
<u>Total Other Expenses</u>	\$205,222
<u>Total Expenses</u>	\$207,622
INDIRECT COSTS (if applicable)	
Federal Agency	
Rate Agency (.0000)	0.15
Base	86956.00
TOTAL INDIRECT COSTS	\$13,043

Project Income

Cash Project Income	
Description	Amount
Autzen Foundation	\$5,000
Oregon Arts Commission	\$10,000
Individual Donations	\$3,665
Other Grants	\$25,000
<u>Total Cash</u>	\$43,665

In-kind Project Income	
Description	Amount
Oregon State University (Subject Matter Expert)	\$19,000
Oregon Health and Science University (Subject Matter Expert)	\$19,000
Eastern Washington University (Subject Matter Expert)	\$19,000
Columbia Regional Inclusive Services (Subject Matter Expert)	\$19,000
Telegraph Mastering (Audio Mastering Engineer)	\$1,000
<u>Total in-kind</u>	\$77,000

TOTAL PROJECT COSTS/EXPENSES (Total Direct Costs + Total Indirect Costs)

\$ 220665.00

Amount Requested from the NEA

\$ 100000.00

Total Cost Share/Match for this project (Total CASH + Total IN-KIND)

\$ 120665.00

TOTAL PROJECT INCOME (Total Cost Share / Match + Amount Requested from NEA)

\$ 220665.00

Additional Project Budget Notes (Optional):

CymaSpace personnel are long-time, part-time staff with no fringe benefits. We have a reliable community of foundation partners as well as organizational partners for this project.

Uploaded Docs, Work Samples

Support Material	Title	Description
NIME24_paper58_GestoLumina [CameraReady].pdf	Sample Model Paper	This research paper reflects a survey instrument developed by the project team to be utilized by the proposed work.
NEA - CymaSpace - Letters of Support - 2025.pdf	Letters of Support	Letters of Support from Project Partners

Organization & Project Data

Applicant Organization Discipline: Select the primary discipline that is most relevant to your organization. This refers to the primary artistic emphasis of your organization. *

- Presenting & Multidisciplinary Works

Additional Applicant Organization Discipline: Optionally, choose up to two additional disciplines

- Artist Community
- Film & Media Arts

Applicant Organization Description: Select the primary description that is most relevant to your organization *

- Artists' Community, Arts Institute, or Camp

Additional Applicant Organization Description: Optionally, choose up to two additional descriptions

- Arts Service Organization
- Media-Film

Does your organization have a disaster preparedness plan (i.e., plans to handle disasters and emergencies such as floods, hurricanes, pandemics, and human-made disasters)? *

- No

Project Activity Type: *

- Curriculum Development/Implementation - Includes design, implementation, distribution of instructional materials, methods, evaluation criteria, goals, objectives, etc.

Additional Project Activity Types: Optionally, choose up to two additional types

- Arts Instruction - Includes lessons, classes and other means to teach knowledge of and/or skills in the arts
- Research/Planning - Includes program evaluation, strategic planning, and establishing partnerships

Race/Ethnicity (choose all that apply)

- No specific racial/ethnic group

Age Ranges (choose all that apply)

- Children/Youth (0-17 years)

Underserved Communities

When thinking about the underserved communities your project intends to serve, please select the factors that limit their opportunities to benefit from arts programming:

- Disability